Health-a-thon 2026

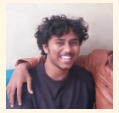
FINAL SUBMISSION

Sahara

A palliative-care companion. Doctor-led goals-of-care conversations, written onto the legal form and ready in an emergency.

TEAM NAME Zeros and Ones

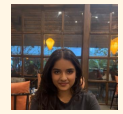
TEAM CODE [add team code]


Shaik Abdul Basith


Shaik Muhammad Awaiz


Dr Sharada Vinod Kutty


Dr Sujay Halkur Shankar


Shirin Ayub

sahara.wedevit.in · Android app and website · English, Hindi, Telugu, Urdu, Kannada

Team Overview & Background



Shaik Abdul Basith
Technical lead

Product design, website, AI (Gemini, Whisper) and the goals-of-care flow.



Shaik Muhammad Awaiz
Technical member

Android app, Supabase server, deployment and phone testing.



Dr Sharada Vinod Kutty
Doctor partner

Pulmonary specialist, lung-cancer care. Wrote the clinical specification: care plan, living will, emergency card.



Dr Sujay Halkur Shankar
Clinician

Pulmonary specialist with palliative-medicine experience. Chose the problem, set the goals-of-care checklist, led app walkthroughs.



Shirin Ayub
Medical student

Symptom tracker: symptom list, scales and check-in wording.

Doctor partners. Both are pulmonary specialists who have worked extensively with lung cancer, and Dr Sujay also has palliative-medicine experience. They chose the problem, wrote the clinical rules, and reviewed every build; the last walkthrough was on 3 Nov.

Legal review. Rituparna P reviewed the goals-of-care form; the app copies it word for word. **Organisation:** independent team; apps published under WeDevit.

Problem Statement and Supporting Evidence



Illustrative scene. A 62-year-old woman with advanced breast cancer.

At the clinic, her doctor discusses her wishes. Only her husband attends.

Two months later, she is unconscious in the Emergency Department. Did she want CPR?

The family disagrees and her earlier note is missing.

15.6 lakh

new cancer cases in India in 2024.

IARC, GLOBOCAN 2024

98.39%

unmet palliative-care need in metastatic cancer.

Patil et al., 2024

88%

of surveyed urban patients wanted to choose their end-of-life treatment.

IndiaSpend, 2019

11.5%

of doctors (45/391) in one hospital were trained in advance directives.

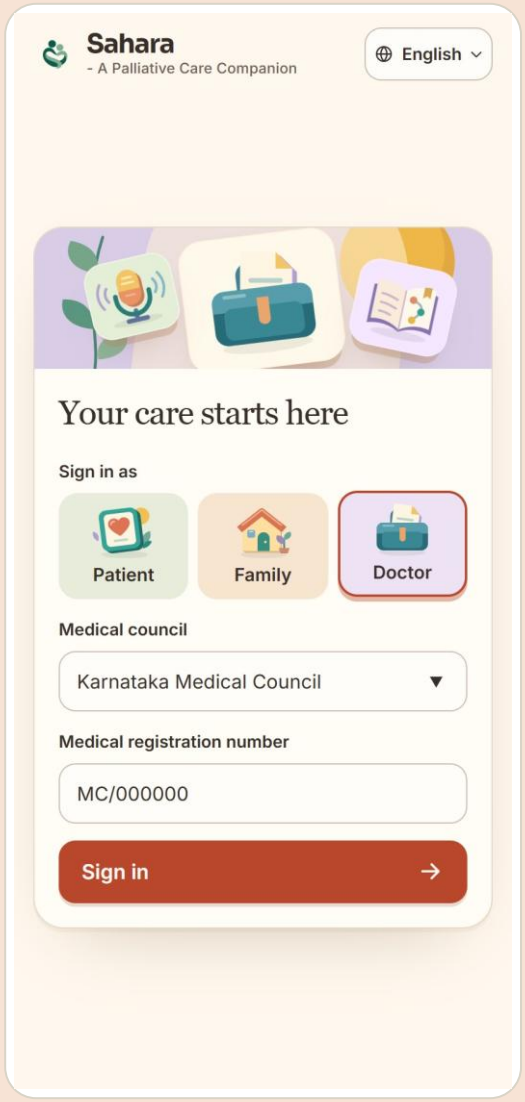
Jacob et al., 2026

What needs to improve

A doctor-led goals-of-care conversation, written down on a legally valid form, signed by doctor and patient or family, and found in seconds in an emergency. Between visits, symptoms and medicines need a simple record the doctor can see.

Studies of different groups; not results from Sahara.

Target Users and Use Case



Patients receiving palliative care At home, between clinic visits.
Record how they feel, keep medicines and reports, and know their wishes are written down.

Family caregivers At home, often deciding for the patient.
See the same record, log symptoms, and read exactly what the patient said, signed.

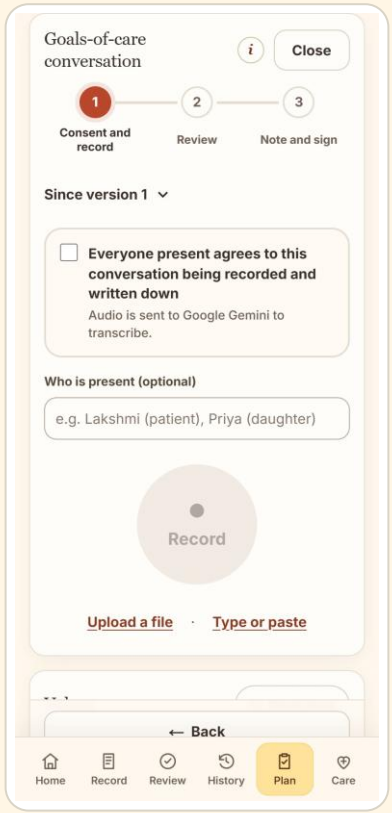
Palliative, oncology and pulmonary doctors Outpatient clinic and ward.
Lead the goals-of-care conversation; the app writes it onto the legal form so they only review and sign.

Emergency doctors Emergency Department, minutes to decide.
Scan the patient's QR card to read their wishes, decision-maker and medicines.

Fits their workflow: one app, three sign-ins (patient, family, doctor); five languages; works on low-cost Android phones.

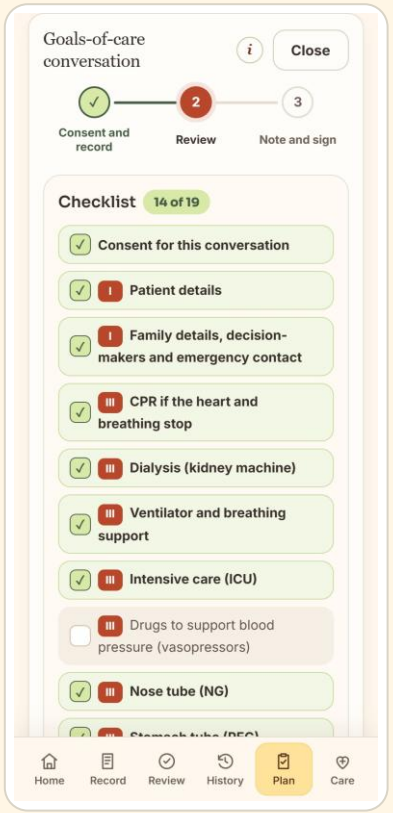
Solution Overview, Key Features & Impact (1/2)

Doctor leads. AI writes it down. The goals-of-care conversation, from consent to a signed note.



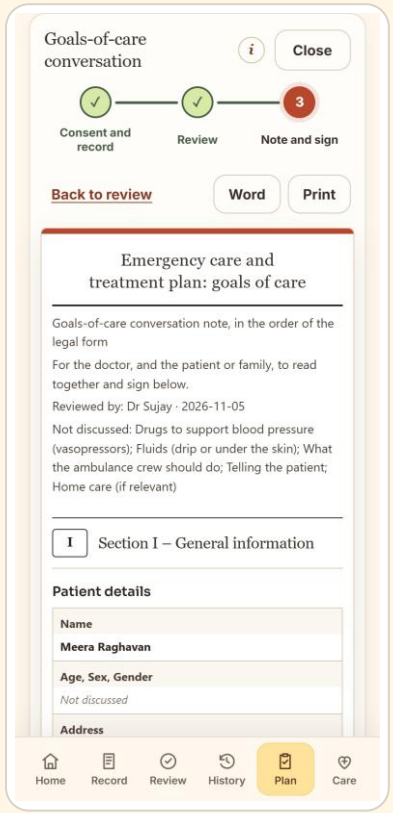
1. Consent, then record

Everyone agrees. Gemini transcribes; Whisper on the phone if offline.



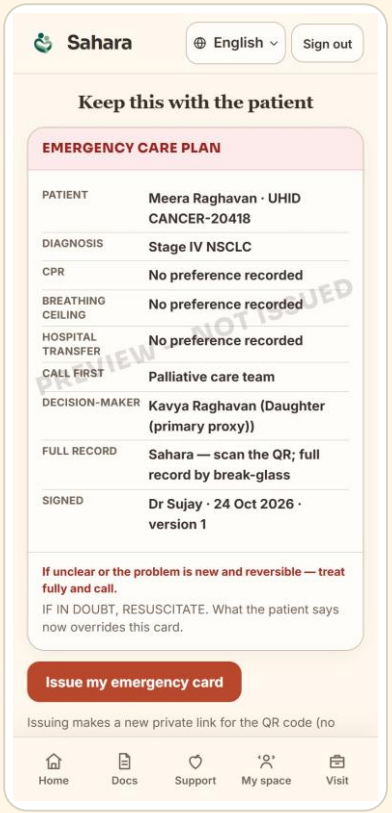
2. Checklist ticks itself

Topics in the doctors' order. Gaps are flagged before the note.



3. Written on the legal form

Sections I–IV, word for word. The doctor reviews and edits.

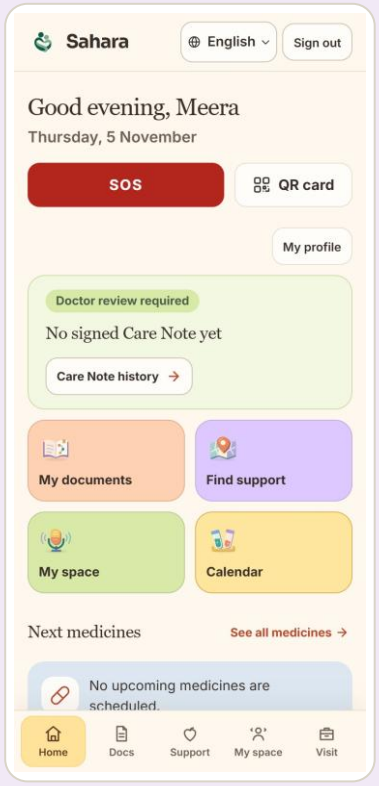


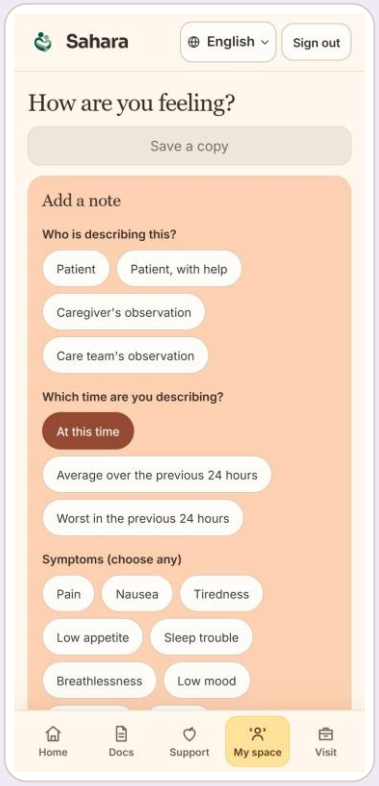
4. Signed, locked, findable

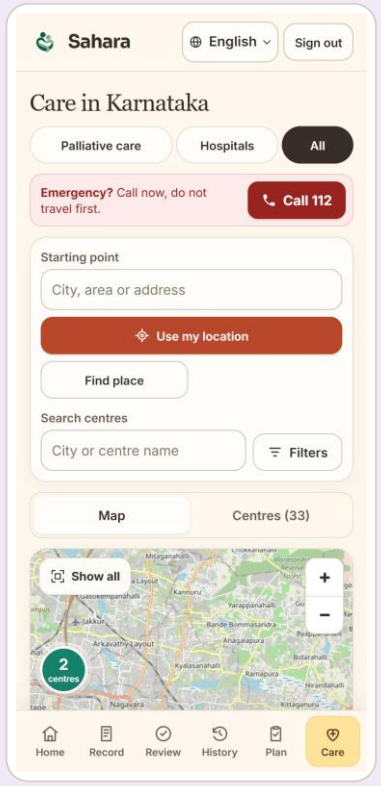
Doctor and patient or family sign. Wishes go on the emergency QR card.

Solution Overview, Key Features & Impact (2/2)

Between visits: a companion for patients and families, shared with their doctor.







How I feel: symptoms, 0–10 distress and a pain body map; trend charts for the doctor.

Medicines: each dose is a tick, with reminders and prescription scan.

Labs and documents: trends, and “Ask your documents” answers with the source line.

Emergency QR card and SOS: wishes, contacts and medicines in one scan.

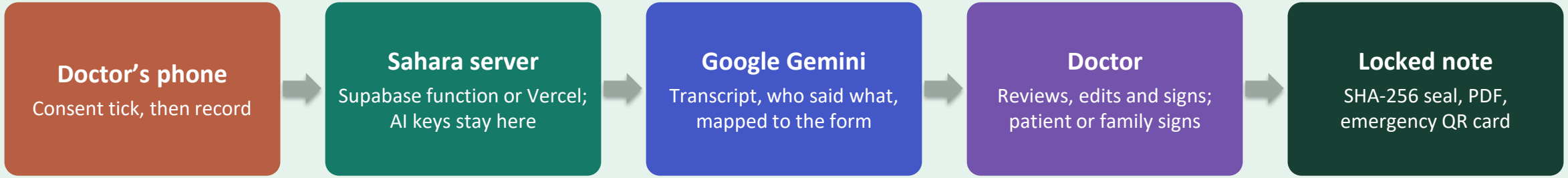
Help nearby: palliative centres, ambulances and support groups.

Family: invited relatives see and add to the same record.

Expected impact (to be measured in the pilot):

more goals-of-care conversations written and signed · wishes found in seconds in an emergency · fewer family disputes at the bedside · symptoms seen before the next visit.

Technology, Tools and AI Approach



If Gemini is busy: next model (Flash-Lite, then Flash) and next key. **Offline or no consent:** OpenAI Whisper on the phone or in the browser, then typing. Speakers are separated on the device.

What AI does

- Turns speech into text (Indian languages).
- Labels speakers.
- Places each statement in the right section of the legal form; flags topics not discussed.
- "Ask your documents": answers with the source line.
- It never diagnoses or suggests treatment.

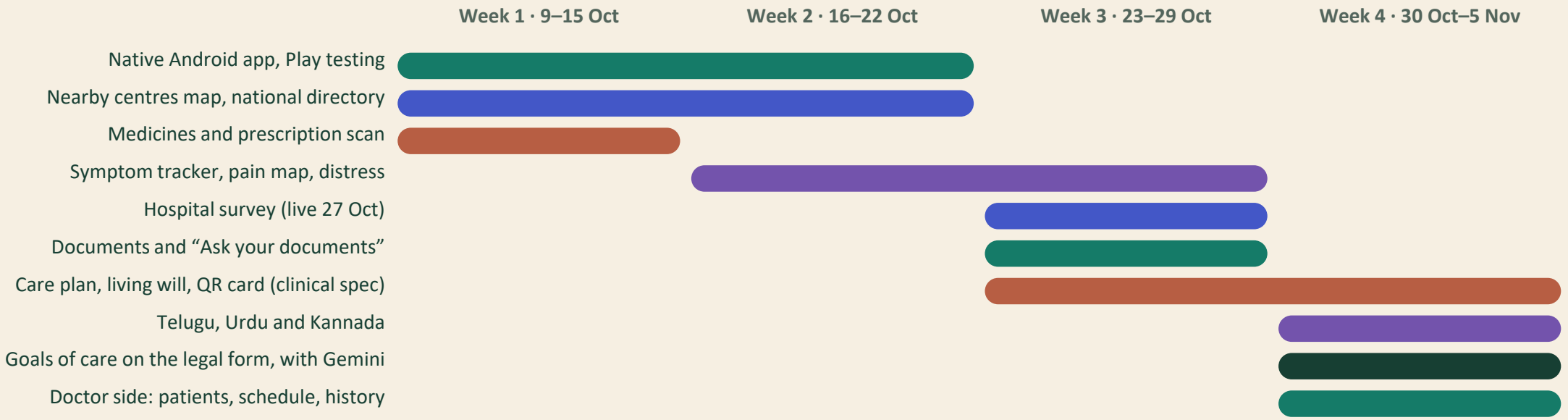
Tools and models

- Google Gemini 3.x Flash-Lite and Flash.
- OpenAI Whisper (on device).
- Google ML Kit translation and read aloud.
- Android: Kotlin, Jetpack Compose.
- Web: Next.js on Vercel.

Data and integrations

- Supabase: sign-in, Postgres with row-level security, Edge Functions.
- Legal goals-of-care form (Rituparna P).
- Clinical spec (Dr Sharada); Vidhi living-will template.
- Pallium India directory and OpenStreetMap.
- FHIR R4 export for hospital systems.

Build Sprint Journey and Progress



Working now

Patient and family app; doctor side; goals of care from recording to a locked, signed note; QR card; website live; Android app on Google Play testing.

How it evolved

Round 1: a web prototype where AI drafted a “Care Note”. Now: a native app; the doctor leads; the note is the legal form word for word; daily companion added from doctors’ walkthroughs.

Challenges and changes

Gemini free tier said “high demand”: added model and key fallback. Legal wording: copied the form exactly. Faded text on some phones: fixed.

Still incomplete

Full test on real phones; a recorded conversation with Dr Sharada; clinician back-translation; open legal questions (capacity, partners).

Evaluation and User Validation

What we tested	How	Result
Code	Automated tests on every build	Android: 877 unit tests pass. Website: 594/594.
Goals-of-care mapping	8 fictional conversations with expected answers: English, Hindi, Hinglish (change of mind), Telugu, Urdu, Kannada, family disagreement, short talk	8/8 matched the expected form.
Speech to text	Gemini in the cloud; Whisper on the device; synthetic audio	English works end to end. Telugu and Kannada weak on the device, so typing is offered.
Ask your documents	4 questions on a practice discharge PDF	4/4 correct, with the source line.
Hospital survey	Real browser flows; changed after doctors' feedback	Live since 27 Oct.

Feedback and what we changed

Dr Sujay's walkthrough, 3 Nov: added tubes-and-lines care, "If…" wording in the care plan, darker text, a shorter side-effect list, fewer ambulance types.

Legal review, 31 Oct: the note now copies the legal form word for word; 31 comments tracked.

Dr Sharada's clinical spec: the 12 "care that always continues" items used word for word.

Limits and what still needs validation

No real-patient pilot yet. Most checks ran on an emulator or in a browser; the full real-phone test is pending. Gemini accuracy on real Hindi, Telugu and other Indian-language audio is not yet measured. Translations are machine-drafted and need clinician back-translation.

Safety and Responsible AI

Privacy, consent and security

- Recording starts only after everyone agrees.
- Audio goes to Gemini only with that consent; otherwise Whisper runs on the phone.
- Per-patient access, by invite; row-level security; AI keys only on the server.
- Patients can download their data.

AI limits and errors

- AI only transcribes and fills the form; it never advises treatment.
- Topics not discussed are listed, not guessed.
- If a person changes their mind, the latest statement wins and the earlier one is shown struck through; disagreements show both.
- Busy or failing model: next model, then the phone, then typing.

Human oversight

- The doctor starts, reviews, edits and signs first.
- Patient or family signs on the same phone.
- The version is locked with a SHA-256 seal; any edit voids the signatures.
- Every version stays in the history.

Bias, accessibility and inclusion

- Five languages, with read aloud.
- Large text; thumb signature for people who cannot write.
- Works on low-cost Android phones.
- Speech weak in Telugu and Kannada: typing offered. Demos use fictional patients only.

Deployment, Cost and Sustainability Plan

To run a pilot

A palliative or oncology clinic with our doctors; Android phones for doctors; 2 short training sessions; a consent and privacy process under India's DPDP Act; the legal team's open questions settled.

Integration

Not required to start: Sahara works on its own. Later: FHIR R4 export into the hospital record, and a link to CARE (Open Healthcare Network) palliative programmes.

Scale and sustain

Free for patients and families. Clinics, hospitals or NGO palliative programmes pay a small monthly fee per clinic; grants or CSR fund the first pilots. New languages are added through the translation dictionary.

Item	One-time	Ongoing
Hosting: Supabase Pro + Vercel Pro	–	≈ ₹3,900 / month
Gemini, per 30-min conversation	–	≈ ₹2 (per transaction)
Clinic set-up and staff training	≈ ₹25,000	–
Consent, privacy and legal review	≈ ₹25,000	–
Support and updates (part-time)	–	≈ ₹10,000 / month
Per clinic, at 10 clinics	**≈ ₹5,000**	**≈ ₹1,600 / month**

Pricing assumptions

US$1 ≈ ₹86. Supabase Pro US$25 and Vercel Pro US$20 a month. Gemini Flash-Lite paid tier, about US$0.30 per million audio tokens; 30 minutes of audio ≈ 58,000 tokens. 100 conversations per clinic a month. *Team estimates, not quotes.*

Suggested price: ₹1,500–2,000 per clinic per month; free for patients.

What Next? Plans to take initiative forward (if any)

1

Finish and prove the app

Real-phone tests; record a goals-of-care conversation with Dr Sharada; clinician back-translation of Hindi, Telugu, Urdu and Kannada.

2

Pilot for 60–90 days

With ethics approval, in a palliative-care clinic. Compare before and after: goals-of-care conversations documented; signed notes per clinic; time to find the patient's wishes in an emergency drill.

3

Settle the legal questions

With Rituparna P: capacity checks, partners as decision-makers, withdrawal of treatment (Common Cause judgments).

4

Fund and grow

Apply for grants and CSR support; offer Sahara to palliative-care networks; link to hospital records through FHIR.

Make the solution information public? Yes.

Our slides, brief, demo video and app description may be shared on Health-a-thon and other websites. This excludes patient records, credentials and the private code repository.

Health-a-thon 2026

Thank you

Zeros and Ones • [add team code]

workwithdevit@gmail.com | sahara.wedevit.in | Demo video: [add demo video link] | github.com/basithladdu/healthathon (private)



Shaik Abdul Basith



Shaik Muhammad Awaiz



Dr Sharada Vinod Kutty



Dr Sujay Halkur Shankar



Shirin Ayub

Health-a-thon 2026